Focus (or) Focal point: Light rays travelling parallel and close to the principal axis when incident on a spherical mirror, converge at a point for concave mirror (or) appear to diverge from a point for convex mirror on the principal axis. This point is called the *focus* (or) *focal point F* of the mirror.

Focal length: The distance between the pole *P* and the focus *F* is called the *focal length f* of the mirror.

Focal plane: The plane through the focus and perpendicular to the principal axis is called the *focal plane* of the mirror.

All the above mentioned terms are shown in Figure 6.7 for both concave and convex mirrors.

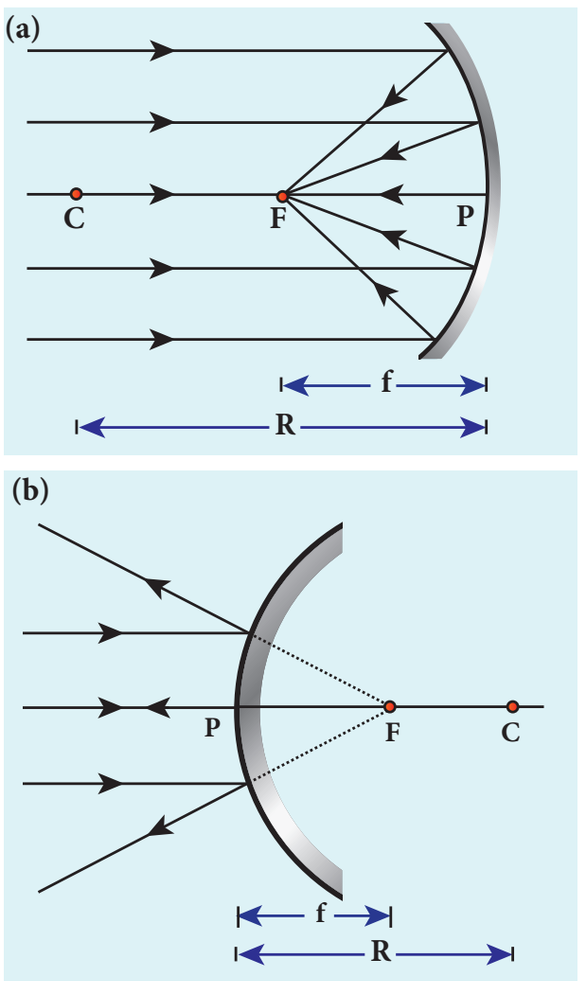


Figure 6.7 Representation of terminologies for (a) concave and (b) convex mirrors

6.2.1 Paraxial Rays and Marginal Rays

The *paraxial rays* are the rays which travel very close to the principal axis and make small angles with it. They fall on the mirror very close to the pole. On the other hand, **the *marginal rays* are the rays which travel far away from the principal axis and make large angles with it**. They fall on the mirror far away from the pole. These two rays behave differently (get focused at different points) as shown in Figure 6.8. In this chapter, we shall restrict our studies only to paraxial rays. As the angles made by the paraxial rays are very small, we can make good approximations.

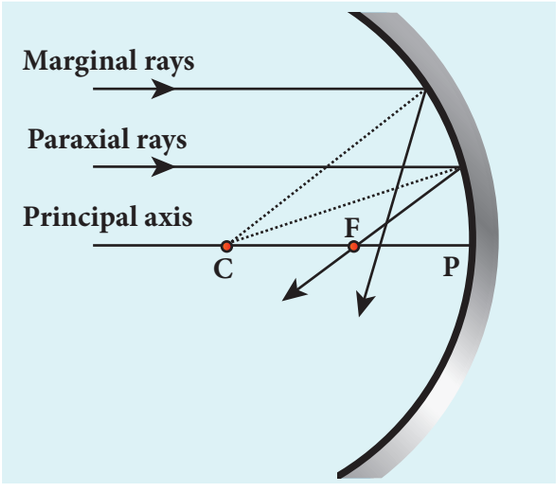


Figure 6.8 Paraxial and marginal rays

6.2.2 Relation between *f* and *R*

Let *C* be the centre of curvature of the mirror. Consider a ray of light parallel to the principal axis is incident on the mirror at *M*. It passes through the principal focus *F* after reflection. The geometry related to this is shown in Figure 6.9(a). The line *CM* is the normal to the mirror at *M*. Let *i* be the angle of incidence and the same will be the angle of reflection.

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